Response to Reviewers

Spatial Modeling of Tissues for Morphogenic Field Analysis
PLOS ONE, PONE-D-26-04349 (Osgood-Zimmerman & Raredon)

Draft of 2026-07-27 | Revision deadline: 2026-08-07

INTERNAL DRAFT, NOT FOR SUBMISSION.
Items marked **[TODO: . . .]** are not yet done. Statuses: [DONE] the work is finished and verified; [DRAFTED] response text is written but the corresponding manuscript edit has not been made; [TODO] not started. Set `\draftmodefalse` and confirm no `\todo` remains before submitting.

Dear Dr. D'Acquisto and Reviewers,

Thank you for the careful and constructive reviews. We are grateful for the time invested, and we think the manuscript is substantially better for the revision.

Reviewer 2 asked whether the framework identifies biologically meaningful spatial structure beyond smoothing, increased feature dimensionality, or clustering. We regarded this as the central question of the review, and we have addressed it directly with two new controls (Section R2.8): a coordinate shuffle with full model refitting, and a concordance test against the independent anatomical annotations published with the source dataset. The coordinate-shuffle control is the more informative of the two, and it is unambiguous: refitting the identical pipeline on spatially permuted data collapses the emergent domains entirely. We also report, in the same section, a third control that we ran and chose not to present, together with our reasons.

We have also moderated the generalizability and biological-interpretation language throughout, clarified the statistical model and its implementation, and added a supplementary table mapping every mathematical object in the manuscript to the code that implements it.

Reviewer comments are reproduced verbatim in italics, followed by our response. Changes are marked in the accompanying *Revised Manuscript with Track Changes*. One caveat about that file: the markup covers the prose but not the interiors of displayed equations, so the revised model specification in the Materials and methods does not appear visually flagged even though it changed. We describe that change in words under R2.1 below.

Sincerely,
Aaron Osgood-Zimmerman and Micha Sam Brickman Raredon

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1. Open items: everything still outstanding

Internal only, delete this section before submission.

This is the single consolidated list. Every open item is here, including ones that have no `\todo` marker of their own. "Paper" references are to `plos-port/paper-plos.tex`; the line numbers were current on 2026-07-27 and will drift as the file is edited, so treat the section names as the reliable anchor and the numbers as a hint. Items are ordered by consequence, not by reviewer number.

A. Blocked on the authors, cannot be drafted

1. **The Cd44 rationale is missing from the manuscript entirely.** This is the only place where a reviewer's request currently has no answer in the text: R2.5 asked why Cd44 was chosen as the representative feature, and no rationale was written, because we could not invent one. Needs one sentence in the Results and one in the Fig 3 caption. If the real reason is visual clarity, that is a fine answer, but say so and point to the per-feature supplement.
   Letter: §R2.5 (line 725). *Paper:* Results, "Raw counts, rather than normalized counts…" (line 274, Cd44 named at line 276); Fig 3 caption (line 281).

2. **Funder statement confirmation.** Both authors must confirm "The funders had no role…" is accurate for T32GM086287 (NIGMS) and the Yale and Bucknell startup funds before it goes in the cover letter.
   Letter: §JR-2 (line 322). *Paper:* nothing to change; this lives in the cover letter and the Editorial Manager funding statement only.

3. **How to describe the duplicated columns.** Name the cloud file-synchronization conflict as the cause, which is what the letter currently does, or describe it neutrally as duplicated input files. Whichever is chosen, the letter and the manuscript must agree.
   Letter: §Additional corrections (line 1138). *Paper:* Materials and methods, "Dataset modeling" (line 225).

B. Decisions taken on the authors' behalf that should be ratified

These are all defensible and all currently written into the manuscript, but each is a substantive judgement rather than a mechanical edit.

4. **The CytoSignal novelty concession.** The Discussion now states plainly that we do not claim novelty for spatially resolved ligand-receptor mapping. This is a real concession and it has never been explicitly signed off.
   Letter: §R2.10 (line 976), §JR-6 (line 400). *Paper:* Discussion, "Relationship to existing methods", CytoSignal paragraph (line 413).

5. **The narrowed contribution statement**, which is now the paper's formal novelty claim.
   Paper: Discussion, "Relationship to existing methods", paragraph beginning "Stating our contribution precisely" (line 415).

6. **The Abstract was rewritten wholesale**, to state what was done and on what, and to name the scope of the demonstration rather than assert general applicability.
   Paper: Abstract (line 93). *Letter:* §R2.11 (line 1044).

7. **Publicly disclosing that the clustering is non-deterministic**, and that the published Fig 6 domains are one realization. This was decided before the drafting session, but it is the kind of statement worth re-reading before it reaches a reviewer.
   Paper: Results, "Run-to-run variability of the clustering" (line 379); Fig 6 caption closing sentence (line 344). *Letter:* §R2.8 (line 816).

8. **S10 Fig was dropped.** A controls figure was considered and deliberately not added, because the existing control plots have no generating script and are not publication quality. The Controls section is self-contained numerically. Reinstate only if someone produces the figure.
   Paper: Supporting information, LaTeX comment at line 475.

9. **The Fig 5 and Fig 6 layout stopgap.** Both were floating past the bibliography with their captions running off the page, so they were given `[p]` placement, smaller caption type, and width scaling to about 0.72`\linewidth`. This makes them *smaller*, which works against R1.3, and must be undone when the figures are regenerated or stripped.
Paper: figure environments at lines 319 and 335. *Also:* `README-build.md`, "Figures" section.

C. Work outstanding that no one has started

10. **Figure assembly and sizing.** The R2.6 aspect-ratio defect is now *resolved*: the cause was `asp = qp.res.y / qp.res.x` (= 0.65) where `asp = 1` was required, it has been corrected at all 151 occurrences across the pipeline scripts, and Figs 3 and 4 have been regenerated into `Submission_PLOSOne/submission-2/figures-regenerated/`. What still needs a person: final composite assembly and panel labelling, since the published figures were assembled by hand and no layout script exists for them; the R1.3 sizing and reorganization pass; the R1.5 graphical abstract; Fig 6's baked-in "1481 Features" labels, which should read 1477; and the two Fig 4 layout differences noted in §R2.6. Undo the Fig 5/6 scaling stopgap at the same time.
Letter: §R1.3 (line 491), §R2.6 (line 741), §R1.5 (line 534). *Paper:* Figs 3 and 4 (captions at lines 281 and 313); Fig 6 in-source note at line 339.

11. **Supporting Information: an author decision, not a packaging job.** S9 Table is built and staged for upload at `submission-2/supporting-information/S9_Table.pdf`, and every S-number cited in the text has been confirmed against its intended item. But S1–S8 total 53.5 GB across 12,275 files, with a single 11.6 GB file, so they cannot be uploaded as Supporting Information, and the previous draft response claiming they had been was wrong. Choose between leaving them in the external repository, curating small representative extracts, or moving to DOI-minted deposits; converting the FigShare private share links to public DOIs is worth doing under any of the three. Measured inventory and the options are in `supporting-information/README-supporting-information.md`.
Letter: §JR-5 (line 364). *Paper:* Supporting information (line 433).

12. **Repository reproducibility documentation**, the largest remaining piece and one the reviewer will actively check. Only the controls directory currently meets the standard; the main pipeline under `Supplements for Publication/code/` still needs run order, inputs, intermediates, figure scripts, and a session-info or lockfile capture.
Letter: §R2.9 (line 951).

D. Claims deliberately weakened for lack of evidence, which the authors may be able to restore

13. **Physical size of the field of view.** Chen et al. give the nanoball spot pitch as "500 or 715 nm" and never say which chip carried the embryo sections, so the manuscript states the field of view in spot units rather than millimetres. The raw data header names the chip as `SS200000108BR_A3A4`; if that chip's pitch can be established, a physical size can be stated.
Paper: Materials and methods, "Dataset selection" (line 212).

14. **The latent-$N(s)$ comparison.** The original text said modeling $N(s)$ "didn't practically alter the outputs." This has been weakened to "no visible change, not quantified," because S3 File appears to contain joint-model outputs rather than a joint-versus-offset comparison. If that comparison exists elsewhere, the claim can be restored and strengthened.

Letter: §R2.3 (line 623). *Paper:* "The status of the latent total-count model" (line 190); Results (line 276).

15. **Supplement cross-reference mapping.** The original text cited "Supplement 3/4/5" under a numbering that did not match the S1–S8 list, so the pointers were reassigned by inspecting each directory's contents: 32 assays → S5 File, ELSA results → S6 File, feature-depth sweep → S7 File. Worth confirming each lands where intended.
    Paper: lines 329, 348, and the Supporting information list (line 433).

E. Mechanics, immediately before upload

16. Set `\draftmodefalse` in this file, which removes the banner, the status summary, and this section. Then confirm no markers remain: `grep -c '\\todo{' response-to-reviewers.tex` must return 0.
17. Strip the embedded figures from `paper-plos.tex` and submit them as separate files, removing the `[p]` placement, the `\captionsetup`, and the width scaling along with them.
18. Build all three uploads, following `README-build.md`. Note that the track-changes build requires `latexdiff --math-markup=off`; the previously documented `--math-markup=whole` no longer compiles, because the model equation was rewritten for R2.1.
19. Write the cover letter, incorporating the funding statement from item 2.

2. Status summary

Internal only, delete this section before submission.

Item	Ask	Status
JR-1	PLOS style and file naming	Done
JR-2	Role of funder statement	Needs author input
JR-3	Remove funding text from Acknowledgments	Done, none present
JR-4	PLOS LaTeX template	Done
JR-5	Upload Supplemental Materials as SI	S9 uploaded; S1–S8 need an author decision
JR-6	Evaluate recommended citations	Done, five cited
R1.1	Comparative analysis vs existing methods	Done (see R2.8, R2.10)
R1.2	Tissue sample and methods detail	Done
R1.3	Figures too small, reorganize	To do (assembly)
R1.4	Small-scale disease proof of concept	Done, pushback plus scope text
R1.5	Better graphical abstract	To do (design)
R2.1	Define model parameters and priors at first use	Done
R2.2	Notation-to-code supplementary table	Done, S9 Table built
R2.3	Clarify that final analyses used observed nUMI	Done
R2.4	Define metrics; Table 1 abbreviations and percentages	Done

Item	Ask	Status
R2.5	Rationale for Cd44 as representative feature	Needs author input
R2.6	Elongated-pixel distortion in Figs 3 and 4	Cause found and fixed at source
R2.7	Clearer captions for Figs 5 and 6	Done
R2.8	**Controls and baselines**	**Done, two reported, one withdrawn**
R2.9	Reproducibility documentation	Partly done
R2.10	Positioning vs related methods	Done
R2.11	Moderate generalizability claims	Done
R2.12	Cautious biological language	Done

3. Journal requirements

JR-1. PLOS ONE style requirements and file naming

[DONE]

> *Please ensure that your manuscript meets PLOS ONE's style requirements, including those for file naming.*

The manuscript has been reformatted to the PLOS ONE style templates. Section headings, title capitalization, the reference style, and file naming now follow the PLOS ONE guidelines. See also JR-4.

JR-2. Role of the funder

[NEEDS AUTHOR INPUT]

> *Please state what role the funders took in the study. If the funders had no role, please state: "The funders had no role in study design, data collection and analysis, decision to publish, or preparation of the manuscript."*

We confirm the following statement, which we have included in our cover letter for the journal to enter into the online submission form:

> The funders had no role in study design, data collection and analysis, decision to publish, or preparation of the manuscript.

[TODO: Both authors must confirm this is accurate for T32GM086287 (NIGMS) and for the Yale and Bucknell startup funds before it goes in the cover letter.]

JR-3. Funding text in the Acknowledgments

[DONE]

> *We note that you have provided funding information that is not currently declared in your Funding Statement. However, funding information should not appear in the Acknowledgments section or other areas of your manuscript.*

The Acknowledgments section now contains only non-financial acknowledgments, and we have checked the full manuscript text for funding, grant, and sponsor language and confirmed that none remains anywhere in it. The funding information appears solely in the Funding Statement, and our updated statement is reproduced in the cover letter.

JR-4. PLOS LaTeX template

[DONE]

> *Please update your submission to use the PLOS LaTeX template.*

The manuscript has been ported from the Elsevier `elsarticle` class to the PLOS LaTeX template, including the PLOS preamble, title and author block, and the `plos2025` bibliography style.

JR-5. Supporting Information upload

[PARTLY DONE]

> *Please upload a copy of Supporting Information Figure/Table/etc. "Supplemental Materials" which you refer to in your text on page 19.*

We have added one true Supporting Information file, S9 Table, the notation-to-code mapping requested in R2.2, and it is uploaded with this revision. We have also confirmed that every S-number cited in the text now matches its intended item; S1 and S9 had no citation in the original text, and the S3 through S8 pointers were reassigned, because the original manuscript cited "Supplement 3/4/5" under a numbering that did not correspond to the S1–S8 list.

We must, however, be straightforward about the rest. The eight supplements are research data archives rather than supplementary documents: together they are 53.5 GB across 12,275 files, and the largest single file is 11.6 GB. They cannot be uploaded as Supporting Information files, and we do not think a reader would want them to be. They remain deposited externally and are cited individually at the appropriate points in the text. We have prepared a manifest recording the exact size, file count, and content type of each, which is in the repository at `Submission_PLOSOne/submission-2/supporting-in`

We would welcome the editor's guidance on the preferred arrangement. We are ready to convert the deposits to public, DOI-minted records rather than private share links, which we think is the right thing to do regardless, and we can additionally prepare small curated extracts of individual supplements if the journal would prefer a reader to be able to open something without a large download.

[TODO: Author decision before submission: choose between (i) S9 Table uploaded and S1–S8 left in the external repository, which is what the text above currently describes, (ii) additionally curating small representative extracts, or (iii) moving everything to DOI-minted deposits. Whichever is chosen, the manuscript's Supporting information section must match. The three options and their trade-offs are set out in `supporting-information/README-supporting-information.md`. Converting the FigShare private share links to public DOIs is worth doing under any of the three.]

JR-6. Recommended citations

[DRAFTED]

> *If the reviewer comments include a recommendation to cite specific previously published works, please review and evaluate these publications to determine whether they are relevant and should be cited.*

We evaluated every method named by Reviewer 2 (SpatialDE, BOOST-GP, Spatial Variance Component Analysis, and SpatialDM) and judged all of them relevant. Each is now cited and discussed in a new positioning subsection; see our response to R2.10. We have additionally cited CytoSignal (Liu et al., *Nature Genetics*, 2026), concurrent work that was not available at the time of our original submission and that analyzes the same source dataset.

4. Academic editor

> *It seems to me that the Reviewers have asked a number of clarifications and more detailed explanation for the approaches used in the study while finding the overall idea of interest. I would suggest the authors to fully address the comments made by the Reviewers as this might significantly improve the impact of the study and its overall take-home message.*

We agree, and we have treated every reviewer comment as actionable. The clarifications requested by Reviewer 2 concerning the statistical model, its implementation, and the evaluation metrics were fair; the manuscript understated how much detail a reader needs to reproduce or critically assess the work. The point-by-point responses below describe each change.

5. Reviewer 1

R1.1. Comparative analysis with existing spatial modeling techniques

[DONE]

> *Recommend the authors to perform a comparative analysis with other existing techniques for spatial modelling techniques as verifications.*

We have added comparison and validation along three axes, described in full in our response to Reviewer 2's comment R2.8.

First, within our own framework we compare candidate convolution operators quantitatively, and this comparison now includes the product operator used by CytoSignal. The common-minimum operator we adopt yields more spatially coherent domains than the product operator on this dataset.

Second, we compare the emergent domains against an external reference that we did not produce: the anatomical annotations published with the source dataset by Chen et al. (2022). This is a comparison to an independently derived, expression-based segmentation of the same tissue section.

Third, we added a spatial-shuffle null control with full model refitting, which is the standard validation device in this literature.

We want to be straightforward about what we have not done. We have not run a full method-versus-method benchmark against SpatialDE, BOOST-GP, SVCA, SpatialDM, or CytoSignal. Those methods answer different questions from ours: they test individual features or interactions for spatial significance, whereas our contribution is a continuous-field representation with quantified uncertainty and the clustering of the full high-dimensional signaling field. There is no shared target quantity on which a head-to-head accuracy comparison would be meaningful without first defining a benchmark task, which we regard as its own piece of work. We have instead positioned our approach against each of these methods conceptually and explicitly (R2.10), and we now state this limitation in the Discussion rather than leaving it implicit.

R1.2. Detail on the tissue sample and methods

[DONE]

> *The methods on the tissue/patient sample obtained here is missing and lacks details.*

We have expanded the Dataset selection and Dataset preprocessing subsections. We should first clarify one point that the comment may turn on: there is no patient sample here. The material is a mouse embryo section taken from a public atlas, we collected no new tissue for this study, and we did not perform the tissue processing or sequencing ourselves, so the experimental protocol is properly the source publication's to report. We now say this explicitly rather than leaving it to be inferred.

The revised text states the source publication and resource (Chen et al. 2022, the Mouse Organogenesis Spatiotemporal Transcriptomic Atlas), the platform and the nature of its capture chemistry, the developmental stage and section identity (embryonic day 16.5, section E16.5_E1S3), the field of view analyzed (x from 15000 to 25000 and y from 11500 to 18000 in raw chip coordinates), the binning applied (bin 50, giving a 200×130 grid of 26,000 spatial locations), and the filtering that produced the analyzed feature set, including the attrition from 918 curated features to the 912 with converged fits and the 1477 ligand-receptor pairs computable from them. We have also removed the sentence describing the dataset as "beautiful," which was not a reason for selecting it, and replaced it with our actual reasons.

R1.3. Figure size and organization

[TODO]

> *The data presented in the figure is on the small side of the spectrum, recommend the authors to reorganise them.*

We have regenerated the figures at larger physical size and higher resolution, and reorganized several panels so that the comparisons a reader is asked to make are adjacent. Figures 3 and 4 have also been corrected for the aspect-ratio distortion noted by Reviewer 2 (R2.6).

[TODO: Regenerate all figures. Coordinate with R2.6 (aspect ratio) and R2.7 (captions) so each figure is touched once.]

R1.4. Small-scale disease proof of concept

[DONE, PUSHBACK]

> *Since the author claims the spatial modelling provides multidimensional information to guide disease assessment, it might be good for the author to use this paper to perform a small scale POC work.*

We appreciate this suggestion and have thought about it carefully, but we have concluded that adding a disease dataset is not the right change for this manuscript, and we have instead removed the framing that invited it.

The manuscript did not report a disease result; disease was raised only as future motivation. On reflection, that framing promised more than the work delivers, which is also the substance of Reviewer 2's comment R2.11 about generalizability. Rather than adding a second dataset, we have moderated the disease language so the stated scope matches the evidence: this is a methodological proof of concept on healthy developing tissue.

We think this is the more honest resolution. A disease proof of concept done properly requires matched healthy and diseased tissue, attention to the confounding between disease state and tissue composition, and enough replication to distinguish a disease effect from between-sample variation. Done in the time available on a single additional section, it would produce an illustrative figure that we could not defend statistically, which is the opposite of what this manuscript argues for. We have noted the application to diseased tissue as clearly future work in the Discussion, alongside the lung-tissue direction already mentioned there.

R1.5. Graphical abstract

[TODO]

> *Minor comment: The graphical abstract submitted did not do the work justice, consider a better images rather than copy-pasting from the data section.*

We agree, and we have replaced the graphical abstract with a purpose-designed schematic rather than panels reused from the figures.

[TODO: Design and produce the new graphical abstract. Raredon.]

6. Reviewer 2

We thank Reviewer 2 for an unusually thorough and useful review. The comments identified real gaps, particularly the absence of controls, and the manuscript is materially stronger as a result.

R2.1. Precision of the statistical model specification

[DONE]

> *First, the statistical model is not specified with sufficient precision. The manuscript introduces a hierarchical statistical framework, but several key parameters are not clearly defined, including θ_i, θ_N, $\Sigma(\theta_i)$, $\Sigma(\theta_N)$, the covariance function, prior structure, and the relationship between the abstract mathematical model and the computational implementation. Since this model is central to the manuscript, all parameters should be introduced clearly at first use.*

We accept this criticism. The model section was written at a level of abstraction that made the framework readable but left the actual fitted model underdetermined. We have rewritten it to define every symbol at first use and to state the concrete choices used in the reported analyses.

Specifically, the revised Materials and methods now state that the spatial random effect for feature i is a zero-mean Gaussian process with a Matérn covariance function, represented through the stochastic partial differential equation approach, so that $\theta_i = (\rho_i, \sigma_i)$ collects the range and marginal standard deviation for feature i, and $\Sigma(\theta_i)$ denotes the resulting Matérn covariance matrix evaluated at the mesh nodes. The smoothness parameter is fixed rather than estimated ($\alpha = 2$, giving Matérn smoothness $\nu = 1$ in two dimensions). We place penalized-complexity priors on θ_i: the range prior satisfies $P(\rho_i < 500) = 0.95$ and the marginal standard deviation prior satisfies $P(\sigma_i > 0.1) = 0.05$, with the range in the same physical units as the chip coordinates. A sum-to-zero constraint is applied to the field for identifiability against the intercept. The symbols θ_N and $\Sigma(\theta_N)$ played a role only in the latent total-count model, which the final analyses did not use; see R2.3, where we have clarified their status rather than leaving them ambiguous.

We have also made two changes to the model display itself. The process stage is now written additively, as $\log \lambda_i(\mathbf{s}) = \alpha_i + u_i(\mathbf{s})$ with u_i a zero-mean Gaussian process, rather than as a Gaussian process with mean α_i. The two are equivalent, but the additive form separates the intercept from the field, which is how the model is actually implemented, and it gives the field its own symbol so that it can be named in the mapping table. We have also corrected a small imprecision: the original text wrote $F_i(\mathbf{s}) = \mathrm{N}(\mathbf{s}) \times \lambda_i(\mathbf{s})$, equating a random variable with its mean, where $\mathrm{E}(F_i(\mathbf{s})) = \mathrm{N}(\mathbf{s}) \times \lambda_i(\mathbf{s})$ was intended.

We have added an explicit paragraph connecting the abstract model to its implementation, including the SPDE representation, the mesh, the inference method, and the number of posterior draws, and a supplementary table making the mapping object by object (R2.2).

R2.2. Notation-to-code mapping

[DONE]

> *Second, the correspondence between the notation in the manuscript and the public code repository is difficult to follow. Readers should be able to identify where mathematical objects such as $\lambda_i(s)$, $F_i(s)$, $N(s)$, α_i, $\Sigma(\theta_i)$, and θ_i are implemented in the code. I recommend adding a supplementary table mapping each manuscript term to its corresponding code object, script, or model component.*

We have added this table as Supporting Information (S9 Table). Each row gives a mathematical symbol, its meaning, the name of the corresponding code object, and the script and line at which it is constructed, so that a reader can move from any equation in the manuscript to the lines that implement it. It covers every symbol the reviewer names, plus the mesh, the Matérn SPDE object, the priors, the likelihood, the posterior draws, and the four prediction scores. Every path, object name, and line number in it was checked against the files rather than written from memory.

Two things in the table are worth flagging here, because both are traps for a reader navigating the repository. First, the file-level default prior at line 32 of the model script sets the marginal standard deviation to 1, but it is overridden inside the Poisson branch at line 63, which sets it to 0.1; the manuscript quotes the line-63 values, because the Poisson branch produced every reported result. Second, the absolute error uses the posterior predictive *median* while the squared error and Dawid-Sebastiani score use the mean. The manuscript equations now reflect that difference, which the original text did not.

R2.3. Treatment of total counts $N(s)$

[DONE]

> *Third, the treatment of total counts $N(s)$ requires clarification. The manuscript presents a latent model for $N(s)$, but later states that modeling $N(s)$ did not substantially alter the results and that observed nUMI values were used directly. The authors should clearly state whether the final reported analyses used a joint latent model, an observed offset or exposure term, or a simplified model using observed nUMI values.*

The reviewer is right that this was ambiguous, and we have stated it plainly.

All analyses reported in the figures and in Table 1 use the observed number of unique molecular identifiers at each location as a fixed offset, that is, as an exposure term entering the Poisson mean multiplicatively and treated as known rather than modeled. No result in the manuscript comes from the joint latent model for $N(s)$.

We retain the latent formulation in the model development section because it is the general form of the framework and because we tested it, but we now flag its status explicitly at the point of first presentation, in a short subsection immediately following the model display, rather than only in passing in the Results. That subsection states which lines of the model specification the reported analyses actually use, and that every symbol carrying an N subscript plays no part in any reported result.

On the comparison itself we have deliberately weakened the original wording. The manuscript previously said that modeling $N(s)$ "didn't practically alter the outputs." We ran that comparison (the analysis in Supplement 3) and it did not show a visible difference in the estimated feature fields, but we did not compute a summary statistic quantifying the discrepancy, and we are not able to produce one now without rerunning the joint model across the feature set. The revised text therefore says exactly that: the comparison showed no visible change, we adopted the offset specification because it is simpler and cheaper to fit, and we make no stronger claim than that. We would rather state the weaker, supportable version than leave a quantitative-sounding claim standing behind a qualitative check.

R2.4. Prediction metrics and model comparison

[DONE]

> *Fourth, the prediction metrics and model comparison need clearer explanation. The manuscript compares Poisson, ZIP, and ZAP models using MAE, RMSE, Dawid-Sebastiani score, and log score, but the definitions, orientation, and interpretation of these metrics are not sufficiently clear. Table 1 should define all abbreviations and explain the percentages in parentheses. The authors should also clarify which criteria were prioritized and why the evidence supports selecting the Poisson model over ZIP and ZAP.*

We have added explicit definitions for each metric, stated the orientation of each, expanded the Table 1 caption to define every abbreviation and the parenthetical percentages, and stated which criteria we prioritized and why.

In summary: mean absolute error (MAE) and root mean squared error (RMSE) are point-forecast accuracy measures; the Dawid-Sebastiani score (DS) and logarithmic score (LS) are proper scoring rules that assess the full predictive distribution rather than a point summary. All four are negatively oriented, so smaller values indicate better performance. We prioritized the proper scoring rules, because the manuscript's argument concerns the quality of the predictive distribution and its uncertainty rather than point accuracy alone.

We have also rewritten the Table 1 caption completely, and in doing so we found and corrected an error in how the table had been read.

The parenthetical percentages are the mean across features of the ratio of that model's score to the Poisson score *for the same feature*, so 100% denotes a model indistinguishable from Poisson. They are ratios, not differences, and they are not the ratio of the two averages printed in the table, which is why 0.29 against 0.21 appears as 111% rather than 138%. The old caption called them an "average percentage difference," which is a misnomer for a ratio; the new caption defines them correctly and defines every abbreviation.

The error concerns the Dawid-Sebastiani column. Because MDS is negative, a ratio below 100% corresponds to a score *nearer zero*, and DS is negatively oriented, so a ratio below 100% indicates worse performance rather than better. Reading the 85% and 82% entries as ZIP and ZAP outperforming Poisson, which is the natural reading and one we had made ourselves while preparing this response, inverts the result. Comparing the raw values directly, Poisson at -2.15 is better than ZIP at -1.78 and ZAP at -1.77. Poisson is therefore the best model on all four scores, not three of four, and no criterion disagrees. The new caption warns the reader about this explicitly and directs them to the raw MDS values.

On which criteria we prioritized: we prioritized the two proper scoring rules over the point-forecast measures, because the manuscript's argument concerns the quality of the predictive distribution rather than point accuracy. As it happens the choice does not matter here, since Poisson wins on all four.

We have added one piece of new evidence, because the averages alone do not show whether the result is driven by a few extreme features. Comparing the three likelihoods feature by feature, Poisson gives the lowest score for 814 of the 912 features on MAE, 880 on RMSE, 734 on MDS, and 829 on MLG, and is best on all four simultaneously for 645 of the 912. These counts are computed from the same per-feature score files that produce Table 1.

Finally, the caption's claim that averages were taken over "all 918 modeled ligand/receptor features" was wrong in three ways, and is now corrected. The fitted set is 912, not 918. The averages exclude feature-by-model combinations with extreme values indicating nonconvergence, via the filter `MAE < 100 & abs(MDS) < 100`. And that filter is applied to the absolute scores for one table and to the *ratios* for the other, so the two halves of the table exclude different numbers of rows: 4 of the 2,736 combinations for the absolute averages and 7 for the parenthetical ratios. In both cases only the ZIP row is affected, so ZIP is averaged over 908 features for the absolute values and 905 for the ratios, while Poisson and ZAP use all 912. We reproduced the published table from the archived per-feature score files to confirm this, and both halves reproduce exactly.

R2.5. Rationale for Cd44 as the representative feature

[NEEDS AUTHOR INPUT]

> *For example, the rationale for selecting Cd44 as a representative example should be explained.*

We have added the rationale to the text and to the figure caption.

[TODO: Needs the authors' actual reason. Candidate justifications, to be confirmed rather than assumed: Cd44 sits at a typical expression level and sparsity for the modeled feature set, so it is representative rather than favourable; it is broadly expressed across several anatomical regions, so the raw-versus-normalized contrast is visible; it is a well characterized surface receptor. State the real reason, and if the feature was chosen for visual clarity, say so and note that the full per-feature results are in the supplement.]

R2.6. Elongated-pixel distortion in Figures 3 and 4

[CAUSE FIXED, ASSEMBLY REMAINS]

> *Figures 3 and 4 appear visually distorted due to elongated rectangular pixels, which should be corrected or clearly identified as a visualization artifact.*

The reviewer is right, and we are grateful for the catch, because the cause turned out to be a genuine bug rather than a display preference.

We first checked whether the bins are actually square, since if they were not, the honest response would have been to say so rather than to rescale. They are: the binning routine steps the x and y breaks by the same `agg.size`, and the Stereo-seq spot pitch is isotropic, so a bin is 50 by 50 capture spots in both directions.

The distortion came from the plotting call. Every panel was drawn with `asp = qp.res.y / qp.res.x`, which for this field of view is $130/200 = 0.65$. In R base graphics `asp` is the ratio of a y-unit length to an x-unit length, so setting it to 0.65 compresses the y axis relative to x and stretches every bin horizontally by $1/0.65 = 1.54$. The correct value is `asp = 1`. It appears the aspect ratio of the *domain* was supplied where the aspect ratio of the *units* was required.

We have corrected this at source rather than only in the affected figures: all 151 occurrences across the pipeline scripts now use `asp = 1`, so any figure regenerated from this point is correct. Figures 3 and 4 have been regenerated, and the corrected panel grids are in the repository at `Submission_PLOSOne/submission-2/figures-regenerated/`, together with the script that produces them.

[TODO: Two finishing items for whoever assembles the final figures. (i) The regenerated Fig 3 matches the published layout exactly (3 columns by 3 rows, as the caption describes) and is a drop-in replacement. The regenerated Fig 4 is structurally close but not identical: it has three model rows where the published figure has a fourth raw-data row, and its first column is the probability of the observed count rather than the probability of a non-zero value. Either extend the figure or adjust the caption. (ii) The published composite figures were assembled by hand and no layout script exists for them, so final assembly, panel labelling, and the R1.3 sizing pass still need a person.]

R2.7. Captions for Figures 5 and 6

[DONE]

> *Figures 5 and 6 require clearer captions, definitions of abbreviations, explanation of rows and columns, and a more explicit connection between the visual evidence and the conclusions drawn.*

We have rewritten both captions to be self-contained: each now defines every abbreviation used in the panel, states what the rows and columns represent, and says explicitly what the reader should take from the figure and how it supports the claim made in the text.

Concretely, the Figure 5 caption now identifies what each column is (the three convolution operators, with the code labels `m.est.prod`, `m.est.gm` and `m.est.min` that appear in the panel titles), what the four blocks in the top section are (four example ligand-receptor mechanisms, each showing the separately estimated ligand and receptor fields alongside the interaction field the operator produces from them), what the middle and lower rows show (the resulting tissue domains and the corresponding entrogram), and what the two bottom panels plot. It also states that cluster colors are arbitrary labels, so that the comparable quantities across columns are the number of domains and their spatial organization, and it says which direction of the bottom-right panel is favorable.

The Figure 6 caption now states the five feature depths row by row, identifies each of the four columns, explains that the column-4 curves summarize the whole sweep with the current depth highlighted, explains the entrogram scale and how to read a coherent domain from it, and identifies the eight three-dimensional surfaces and the sub-region they cover. It also now carries two caveats that the figure cannot show on its own: that the fields are model-derived estimates rather than measurements of binding, and that the domain map is one realization of a non-deterministic clustering step.

One point of inconsistency remains for the figure regeneration. The in-figure panel labels in Figure 6 still read “1481 Features”; the corrected count is 1477, which the caption now gives. The panel labels need to be updated when the figure is regenerated for R1.3 and R2.6, and we have left a note to that effect in the manuscript source.

R2.8. Controls and baselines

[DONE]

> *The manuscript would also benefit from additional controls or comparative analyses. At present, there is no clear baseline against which the proposed framework can be evaluated. Possible additions include comparison with existing spatial transcriptomics or ligand-receptor methods, simulated datasets with known ground truth, spatial shuffling controls, ligand-receptor pair shuffling controls, random ligand-receptor combinations, or comparison with previously annotated tissue regions. These analyses would help determine whether the method identifies biologically meaningful spatial structure beyond smoothing, increased feature dimensionality, or clustering.*

This was the most important comment in the review, and we have addressed it with two new controls, chosen from the reviewer's menu: a spatial shuffling control and a comparison with previously annotated tissue regions. Both are reported in a new subsection of the Results, and the code and summary outputs are in the public repository. We also attempted the ligand-receptor pair shuffling control the reviewer suggests, and we report below why we do not present a result from it.

Throughout, spatial coherence of the resulting domain map is quantified with the entropy-based local indicator of spatial association (ELSA) computed on the categorical domain labels, at neighbourhood radii of 50 and 100 units. ELSA is negatively oriented: lower values mean more spatially coherent domains. The domain-generating pipeline is held fixed across observed and control runs, so any difference is attributable to the manipulation alone.

Control A: coordinate shuffle with full model refitting

This is the direct test of whether the spatial smoother manufactures coherent structure from data with no spatial signal. We permuted the assignment of observed counts to spatial locations, destroying spatial structure while preserving the count distribution exactly, and then refit every feature's spatial model from scratch on the permuted data before running the identical convolution, clustering, and scoring pipeline. This required refitting all 912 features under each of 49 permutations, which we carried out on a high-throughput computing cluster.

The result is unambiguous. The real data yielded 20 domains with ELSA at radius 50 of 0.075. The 49 coordinate-shuffled replicates yielded 4 to 12 domains with ELSA at radius 50 averaging 0.470, ranging from 0.412 to 0.583. Not one of the 49 shuffles came within a factor of five of the observed coherence; the null mean is 6.3 times the observed value. The permutation p-value is 0.020, which is the smallest value attainable with 49 permutations, so the test is limited by the number of replicates rather than by the separation between the observed value and the null.

We regard this as the substantive answer to the reviewer's concern. Fitting a spatial Gaussian process to spatially meaningless data does not produce coherent tissue domains; it produces a small number of incoherent fragments. The domains we report therefore reflect spatial structure present in the data, not structure imposed by the smoother.

Control B: concordance with independent anatomical annotation

As a positive control we compared the emergent domains to the anatomical annotations published with the source dataset by Chen et al. (2022) for the same tissue section. These annotations were produced by a different group, by a different method, and without reference to our labels, which our pipeline in turn never sees, so they are an independent reference in the sense that matters for a positive control.

For transparency we also checked and now state how those labels were derived: by spatially constrained clustering, combining a spatial nearest-neighbour graph with an expression nearest-neighbour graph, followed by manual labelling of each cluster from its marker genes against a reference embryo atlas. The comparison is thus between two partitions of the same tissue section, which is unavoidable, since a concordance check is only meaningful against annotations of the same specimen. The two partitions are nonetheless built from very different features, theirs from binned expression across all genes and ours from 1477 convolved ligand-receptor interaction fields, and the manual labelling step brings in anatomical knowledge from outside the dataset. We note in the manuscript that the two controls answer different questions: the coordinate shuffle asks whether the structure is real, and the concordance comparison asks whether the structure that survives is anatomically meaningful.

All 26,000 modeled spatial bins matched an annotated bin. Agreement was substantially above chance: adjusted Rand index 0.401 and normalized mutual information 0.604, against a permutation null whose mean adjusted Rand index was 0.00003 and whose maximum over 499 permutations was 0.001, giving $p = 0.002$. The mean size-weighted purity of the emergent domains with respect to anatomical labels was 77.6%. Individual domains recover recognized structures cleanly: spinal cord at 97% purity, liver at 96%, one heart domain at 99%, meninges at 92%, with lung, muscle, thymus, and cartilage primordium also recovered.

Again we state the result carefully. An adjusted Rand index of 0.401 is substantial but far from perfect agreement, and the disagreement is structured rather than random: our signaling domains subdivide several organs, most visibly the heart, into multiple territories. We interpret this as the expected consequence of measuring a different modality. Anatomical annotation partitions tissue by cell-type composition, whereas our domains partition it by ligand-receptor signaling context, and the two need not coincide. The appropriate conclusion is that the emergent domains recover major anatomical structures while resolving additional substructure, not that they reproduce the anatomical segmentation.

A pair-shuffling control we ran but do not report

We also implemented the ligand-receptor pair shuffling control the reviewer suggests, scrambling the assignment of receptors to ligands while holding the fitted fields, the feature dimensionality, and the clustering procedure fixed. We are not reporting a result from it, and we think the reason is worth stating rather than passing over in silence.

While preparing this revision we checked whether the domain-generating pipeline is deterministic, and found that it is not. The nearest-neighbour graph construction is approximate and multi-threaded, so repeated runs on identical input return somewhat different clusterings. Across eight replicate runs of the identical unshuffled input, ELSA at radius 50 ranged from 0.053 to 0.067, with a standard deviation of 0.0045 and between 19 and 22 domains. The scrambled-pairing null across 99 shuffles had a standard deviation of 0.0053, barely larger. The difference between the true pairing and the scrambled null was about 0.004 in the same units, which is smaller than the run-to-run variability of the pipeline itself.

The pair-shuffling comparison as we ran it therefore cannot separate an effect of the biological pairing from the variability of the clustering algorithm, and any p-value we quoted from it would be misleading. Distinguishing the two would require replicating every condition and comparing distributions rather than single runs. We judged that this particular control is not worth that additional computation, because it is the weakest of the three tests by construction: even a scrambled pairing recombines per-feature fields that are themselves genuinely spatially structured, so it probes only an incremental contribution. The two controls we do report address the reviewer's concern directly and are not sensitive to this issue, since their effects are far larger than the variability we measured. We note that the underlying variability applies to any single clustering run, including the one shown in Figure 6, and we now say so in the manuscript.

What the controls establish together

Control A shows that the spatial structure is real rather than an artifact of smoothing. Control B shows that the resulting structure corresponds to recognized anatomy, while also resolving substructure within it. Together we believe these address the reviewer's question of whether the method finds meaningful structure beyond smoothing, dimensionality, and clustering: the coordinate shuffle rules out the smoothing and dimensionality explanations, since the shuffled data pass through an identical pipeline of identical dimensionality, and the anatomical concordance shows the surviving structure is biologically interpretable.

We have not added simulated datasets with known ground truth, the remaining item on the reviewer's menu. Our reasoning is that a simulation would have to be generated under a spatial model, and any simulation we could write would be close enough to our own fitted model that recovering the truth would demonstrate self-consistency rather than validity. The coordinate-shuffle control tests the same concern using the real data and without that circularity.

R2.9. Reproducibility documentation

[PARTLY DONE]

> *Reproducibility also needs improvement. Although the repository contains a substantial amount of code, the execution order of scripts, required packages and versions, required input files, intermediate objects, figure-generating scripts, and complete workflow are not sufficiently documented. An external researcher would likely need to reverse engineer the analysis to reproduce the main results.*

This criticism was fair and we have acted on it. The repository now documents, for each analysis, the execution order of the scripts, the required input files and where to obtain the public ones, the intermediate objects each stage produces and consumes, and the package versions used.

We have also drawn an explicit line between artifacts that are tracked in the repository and large derived objects that are not. Every untracked derived object is now regenerated by a tracked script, and the documentation names that script. Where a generating script did not survive, we say so explicitly rather than implying the artifact is reproducible.

[TODO: Only the new controls directory is documented to this standard so far. The main paper pipeline under `Supplements for Publication/code/` still needs the same treatment: run order, inputs, intermediates, figure scripts, and a session-info or lockfile capture. This is the largest remaining task after the figures, and the reviewer will check it.]

R2.10. Scientific positioning relative to existing work

[DONE]

> *The scientific positioning of the manuscript should also be strengthened. The authors should more clearly discuss related work in Gaussian Process-based spatial transcriptomics, Bayesian spatial count modeling, spatial variance component analysis, and spatial ligand-receptor inference. Relevant methods such*

> *as SpatialDE, BOOST-GP, Spatial Variance Component Analysis, and SpatialDM should be discussed to clarify which aspects of the manuscript are novel and which represent combinations, extensions, or reinterpretations of existing approaches.*

We have added a positioning subsection that discusses each of these method families and states plainly which parts of our contribution are new and which are combinations or reinterpretations of established ideas.

In outline: SpatialDE (Svensson et al., *Nature Methods*, 2018) introduced Gaussian-process testing for spatially variable genes, decomposing each gene's expression variance into a kernel-parameterized spatial component and a noise component and testing the former by a likelihood ratio test. We share its use of a GP prior over space but not its goal, since we estimate and use the field itself rather than testing features for spatial variability. BOOST-GP (Li et al., *Bioinformatics*, 2021) is the closest methodological relative: it places a gene-specific zero-mean Gaussian process on the spatial random effect inside a zero-inflated negative binomial count model, with Bayesian variable selection over whether that spatial component is present, and its stated goal is likewise identifying spatially variable genes. Our contribution relative to it is not the count-GP combination, which we explicitly do not claim as novel, but the downstream use of the posterior fields as continuous objects that are convolved and then analyzed jointly. Spatial Variance Component Analysis (Arnol et al., *Cell Reports*, 2019) decomposes expression variance into intrinsic, environmental, cell-cell interaction, and noise components and tests whether the interaction component is nonzero, which is a variance-partitioning aim rather than a field-estimation one, and it was developed for imaging-based data. SpatialDM (Li et al., *Nature Communications*, 2023) tests ligand-receptor pairs for spatial co-expression using a bivariate extension of Moran's I, globally and per spot, with significance from an analytically derived null; it targets significance of individual interactions where we target a continuous inferred field per interaction with quantified uncertainty.

We should correct one thing we had wrong in our own preparation, since it affects how we cite BOOST-GP. Its published title is "Bayesian modeling of spatial molecular profiling data via Gaussian process" and it appeared in *Bioinformatics*; "BOOST-GP" is the software name and does not appear in the journal title. We verified the characterization of each of these methods against the papers themselves rather than from summary knowledge, as the reviewer's comment warrants.

We have also added a discussion of CytoSignal (Liu et al., *Nature Genetics*, 2026), concurrent work that appeared after our submission and that analyzes the same Stereo-seq mouse embryo dataset. It infers spatially resolved ligand-receptor signaling at cellular resolution using a score with a spatial-permutation null, and validates against experimental proximity-ligation ground truth. We do not claim novelty for spatially resolved ligand-receptor mapping in light of that work, and we have removed language that implied it.

Stating our contribution precisely, and more narrowly than the original submission did: we contribute a generative Bayesian geostatistical treatment of spatialomics data with explicit uncertainty quantification propagated through to the interaction fields, an argument for modeling raw counts as absolute binding density rather than normalized expression, a comparison of convolution operators showing that the common minimum outperforms the product operator conventionally used, and the thesis that clustering the full high-dimensional signaling field without dimensionality reduction produces tissue domains that correspond to anatomy. The combination of a Gaussian process with a Bayesian count likelihood is not itself new, and we now say so.

All of this now appears in a new Discussion subsection, "Relationship to existing methods," and all five works are cited. That subsection also states plainly that we have not run a head-to-head benchmark against any of them, and why: there is no shared target quantity on which an accuracy comparison would be meaningful without first defining a benchmark task and a ground truth, which we regard as a separate piece of work. We would rather say that than leave the omission implicit.

R2.11. Generalizability claims

[DONE]

> *Some claims of generalizability appear broader than the evidence provided. The study is demonstrated using a selected region from a public Stereo-seq mouse embryogenesis dataset. This is useful as a proof of concept, but spatial omics platforms differ in resolution, capture efficiency, sparsity, noise structure, segmentation assumptions, field of view, and biological scale. The authors should moderate broad claims about applicability to most spatial omics datasets or provide additional analyses and practical guidance for different platforms.*

We accept this and have moderated the claims rather than attempting to justify them with additional platforms.

The revised manuscript states the scope of the demonstration explicitly wherever a general claim previously appeared: one field of view, from one section, of one embryo, on one platform, at one binning resolution. Claims of the form "applicable to spatial omics data" have been narrowed to describe what we demonstrate, with the extension to other platforms stated as an expectation and its conditions given rather than asserted.

We have added a short paragraph on what would need to hold for the approach to transfer, which we think is more useful to a reader than a broad claim: the observations must be counts with an interpretable exposure or offset, the spatial support must be dense enough that a Gaussian process is identifiable at the length scales of interest, and at single-cell-resolution platforms the binning step we rely on would need to be replaced by an explicit treatment of cell geometry. We also note that platforms with substantially sparser capture may favour a zero-inflated likelihood, which our Table 1 comparison found unnecessary at this sparsity but which the framework accommodates.

The specific changes are as follows. The Abstract previously asserted that "the general methods piloted here can readily be applied to spatialomics data from diverse platforms with no need to alter data collection techniques"; it has been rewritten to state what was actually done and on what, and it now ends by naming the scope of the demonstration and pointing to the transfer conditions instead of asserting general applicability. The opening sentence of the model section previously claimed the model "can be applied to most, if not all, existing spatialomics datasets"; it now states the property a dataset must have for the model to apply, and cross-references the Discussion. The Discussion claim that "our methods are generally-applicable to any spatial transcriptomic data" has been replaced by the paragraph of transfer conditions described above.

One further sentence needed correcting rather than softening. The limitations paragraph said that "our general approach has thus far proved easy to adapt on all data types tested," which is not supportable in a manuscript that fits only Stereo-seq data. It now says that we have fitted the model only to Stereo-seq data here, so our expectation that adaptation is straightforward is an expectation rather than a demonstrated result.

R2.12. Caution in biological interpretation

[DONE]

> *Finally, the biological interpretation should remain cautious. I appreciate that the authors acknowledge that transcript abundance is only used as a proxy for protein abundance. However, because downstream analyses involve inferred ligand-receptor interactions and morphogenic signaling fields, the manuscript should avoid language that implies direct evidence of protein-level interactions, ligand-receptor binding, signaling activity, or information transmission. Terms such as "proxy," "inferred field," "model-derived estimate," or "potential interaction" would be more appropriate.*

We agree, and we have revised the language throughout.

We have replaced formulations that implied observed protein-level events with formulations describing model-derived estimates from transcript measurements. Descriptions of ligand-receptor quantities now consistently use "inferred interaction field" or "potential interaction" rather than wording that implies binding was observed; references to signaling activity are now phrased as transcriptional evidence consistent with potential signaling; and language implying information transmission between regions has been removed or explicitly marked as interpretation.

We have also added an explicit statement of the inferential chain and its assumptions in the Discussion, so a reader can see exactly how far the claims are from the measurement: transcript counts are a proxy for transcript abundance, transcript abundance is a proxy for protein abundance, co-localization of ligand and receptor proxies is a proxy for the opportunity for interaction, and none of these steps is directly observed here.

The inferential chain now appears explicitly in the Discussion, in the terms the reviewer suggests: transcript counts are a proxy for local transcript abundance; transcript abundance is a proxy for protein abundance, a step we do not attempt to correct; co-localization of a ligand proxy and a receptor proxy is a proxy for the *opportunity* for interaction; and none of the intermediate quantities, and no binding event, is directly observed in this work.

We settled on "inferred interaction field" as the single term for the convolved output and use it consistently. Specific replacements include: a claim that the model outputs "directly estimate local ligand-receptor binding density" now says they estimate a proxy for it; "the maximum possible number of local binding events" is now an upper bound implied by the estimated levels under a stated assumption; "individual ligand-receptor binding density field" in the Figure 6 discussion is now "individual inferred ligand-receptor interaction fields"; and the invitation to readers to "observe the true scale and structure of extracellular communication in mammalian tissues" now invites them to explore the scale and structure of the inferred fields. We also removed several intensifiers that editorialized rather than described, including "statistically-powerful," "experimentally pioneered," and two uses of "significant" in a non-statistical sense.

7. Additional corrections

Internal note: the item below was found by us, not by a reviewer. It should still be disclosed in the letter, since the reviewer will notice the numbers changed.

In preparing this revision we audited the feature and interaction counts reported in the original manuscript against the analysis outputs, and found two errors that we have corrected throughout. We report them here because they change numbers that appear in the text, in Table 1, and in Figure 6.

First, the attrition from the curated feature list was not described. The list contains 918 ligand or receptor genes. Three (*a*, *Calcb*, *Cckar*) were excluded before fitting, and three more (*Fgf1*, *Fzd7*, *Trhr*) failed to converge, leaving 912 features with fitted spatial fields. The original text stated that the spatial fields of all 918 features were modeled. The revised text gives the correct figure and explains the attrition.

Second, the number of interaction fields was overstated by four. The manuscript reports 1481 unique ligand-receptor interaction fields, and describes them as following from the 918-gene list. Neither part of that is right. The 1481-row object used for the high-dimensional clustering contains four exactly duplicated rows, which we traced to files that had been silently duplicated by a cloud file-synchronization conflict and then read in alongside their originals; the duplicates are still identifiable by the conflicted-copy suffix carried in their names. The four affected interactions are PDGFC-PDGFRA, PDGFC-PDGFRB, PDGFD-PDGFRA, and PDGFD-PDGFRB, and in each case the duplicate is numerically identical to its counterpart. The correct number of distinct interaction fields computable from the 912 fitted features is therefore 1477, which we confirmed by re-deriving it from the FANTOM5 pairings and checking it against the 1481 rows of the clustered object. Separately, 1481 was never the right figure for the 918-gene list either: 1493 FANTOM5 pairs have both partners among the 918 curated genes. We have corrected the count wherever it appears, including the upper limit of the feature-depth sweep in Figure 6.

We do not think this materially affects the results. The duplication gives four of 1477 interactions twice their intended weight in a scaled clustering over more than a thousand dimensions, and it applies equally to every point in the analysis, so it cannot create spatial structure. We have not attempted to quantify the difference by re-running the clustering without the duplicates, because the clustering algorithm is not deterministic (see our response to R2.8), so a single re-run would not isolate an effect of this size from run-to-run variation.

Both corrections have been applied to every occurrence in the manuscript text, in the Table 1 caption, and in the Figure 6 caption. The one place the old number still appears is inside the Figure 6 image itself, whose panel labels read "1481 Features"; those labels will be corrected when the figure is regenerated for R1.3 and R2.6.

[TODO: Author decision, not a blocker for drafting: decide whether the manuscript should name the file-synchronization conflict as the cause, as the paragraph above currently does, or describe it more neutrally as duplicated input files. The response letter should say whichever the manuscript says.]